

Supplementary Material

Table S1: Reasoning, planning, and memory techniques in LLM-based agents across single-agent, multi-agent, and widely-used categories

Reasoning techniques in LLM-based agents	
Category	Technique/Framework
Single-agent reasoning	TOOLLLM depth-first search-based decision tree [17], EASYTOOL task decomposition [18], MATRIX simulation-based self-critique [48], ToolEmu risk-prompted reasoning [56], ToT [15], CogMir social-context prompts [53], Moral regret mechanisms [75], SHAP-based interpretability [28], Beneficial hallucination-driven hypothesis testing [64], ChatNT perplexity-informed confidence estimation [21]
Multi-agent reasoning	Prompt-Structured Strategic Reasoning [55], AGENTVERSE multi-agent collaborative reasoning [4], Theory-of-Mind (ToM) inspired belief module [52]
Widely used techniques	CoT [3, 9, 12, 13, 15, 23, 26, 27, 34, 35, 39, 52, 53, 56, 62, 66, 74], ReAct [1, 46, 50, 58, 59], Self-Reflection and Self-Critique [1, 2, 21, 24, 27, 41, 43, 46, 53, 54, 57, 62, 67, 74, 78], Self-Consistency methods [8, 19], Multi-Agent Collaborative Reasoning [2, 7, 32, 37], Ensemble and consensus strategies [8, 10, 49]
Planning techniques in LLM-based agents	
Category	Technique/Framework
Single-agent planning	TIMEARENA heuristic planning [46], ReHAC multi-step planning [50], Task stack-based planning [13], Single-shot planning [69], World Knowledge Model guided planning [16], RAFA multi-step trajectory generation [76]
Multi-agent planning	ProAgent modularized planning [3], Plan-first execution paradigm [44], DeCoAgent few-shot prompt-based task decomposition [6], Causal strategy learning systems [70], Star-Agents evolutionary strategy [11], UrbanKGent fixed pipeline [43], SMART-LLM coalition formation [25], Human-in-the-loop workflow [41]
Widely used techniques	Multi-step planning [25, 50, 69, 76], Task decomposition planning [6, 18, 46, 65], ReAct-based planning [46, 58, 73], Reflexion planning [46, 50, 73], Re-planning and dynamic adaptation [63, 67], Plan generation and evaluation [47, 61]
Memory techniques in LLM-based agents	
Category	Technique/Framework
Single-agent memory	RE-Bench [45], RCAgent’s OBSK [19], Memory buffers [60], VLN-Imagine [79], ToolEmu prompt context tracking [56], MLAGentBench compressed Research Logs [61], Formal Reasoning Graphs [15], STEVE-EYE episodic re-prompting [22], Moral alignment prompts [75], AGENTBENCH extended histories [77], CogAgent external screenshots and latent weights [23], Rolling logs for robots [51], Prior tool outputs in context [62], Negotiated ontologies [28], Concatenated dialogue contexts [67]
Multi-agent memory	ProAgent recent-K and relevant-K retrieval [3], ColaCare retrieval-augmented generation [36], Satellite network knowledge bases [27], Re-feeding prior dialogue turns [53], Discounted back-tracking [5], MATRIX running textual memory [48], DeCoAgent persistent long-term storage [6], CompeteAI daybook and comment histories [42]
Widely used techniques	Context window memory [20, 25, 26, 31, 33, 40, 43, 63, 71–73, 78], Conversational and dialogue history [29, 32, 37, 40, 41, 47, 65], Reflexion-style memory systems [1, 30, 50, 57], RAG methods [6, 20, 36, 41], Working memory and scratchpad [15, 30, 69], Long-term episodic memory [31, 68], Short-term memory optimization [34, 38, 40, 57]

Table S2: Identified research gaps and potential research paths for advancing LLM-based agent systems.

Research Gaps	Potential Research Paths
Lack of Verifiable Reasoning	1. Replace unstructured chain-of-thought with logically consistent and auditable reasoning frameworks. 2. Integrate symbolic AI approaches (e.g., BDI agents) for verifiable decision-making. 3. Ground reasoning in domain-specific processes for technical and scientific accuracy. 4. Incorporate fine-grained error detection and correction mechanisms during inference.
Limited Self-Improvement	1. Employ introspection-based reinforcement learning for continuous error identification and correction. 2. Use cooperative multi-agent evolution to share and refine learned strategies. 3. Maintain adaptive capabilities without requiring full-scale retraining. 4. Leverage reflective feedback loops to iteratively enhance performance.
Infrastructure Bottlenecks	1. Optimize inference latency through KV-caching and low-latency decoding pipelines. 2. Implement lightweight, modular architectures for resource-constrained settings. 3. Enable efficient on-device and streaming computation for real-time tasks.
Weak Multi-Agent Communication	1. Design adaptive dialogue protocols inspired by human social cognition. 2. Integrate contextual modeling for dynamic, multi-agent negotiation and coordination. 3. Improve real-time collaboration in competitive and cooperative environments.
Limited Context-Sensitive Collaboration	1. Develop proactive goal refinement methods for dynamic user needs. 2. Enhance co-creative interaction to anticipate and adapt to evolving task requirements. 3. Move beyond passive instruction-following toward proactive task engagement.
Insufficient Personalization	1. Build persistent, evolving user models based on long-term interactions. 2. Incorporate privacy-preserving memory architectures. 3. Adapt multi-modal interfaces dynamically to user preferences.
Vulnerability to Adversarial Triggers	1. Establish robust detection and prevention mechanisms against backdoor attacks triggered by contextual or environmental signals. 2. Implement adversarial robustness testing during training and deployment. 3. Continuously monitor and respond to unsafe emergent behaviors.
Inadequate Explainability	1. Embed explainability directly into the reasoning process. 2. Provide step-by-step justifications for decisions in human-readable form. 3. Tailor explanations to different levels of user expertise.
Incomplete Evaluation Frameworks	1. Establish standardized benchmarks for multi-agent reasoning in complex contexts. 2. Include negotiation-heavy, ambiguous, and competitive tasks in evaluation. 3. Measure adaptability, reliability, and fairness across diverse scenarios.
Neglected Tool Adaptation	1. Treat agent tools as learnable and optimizable parameters. 2. Automate prompt engineering for task-specific and tool-specific optimization. 3. Evaluate the effect of tool adaptation on performance across varied tasks.

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